

Pablo Martínez-Martín

An introduction to the concept of “quality of life in Parkinson’s disease”

P. Martínez-Martín
Department of Neurology,
University Hospital of Getafe,
Carretera de Toledo,
E-28905 Getafe/Madrid, Spain
Fax: 34 1 683 9748
e-mail: pmartinezm@meditex.es

Abstract The sense of well-being is central to the concept of quality of life (QoL), but there is not a universally accepted definition of health-related QoL. QoL refers to the patient’s own evaluation of the impact of the disease. It includes physical, psychological and socioeconomic dimensions, and its measurement is of paramount importance in evaluating research outcomes and in cost-benefit analyses. In addition, QoL assessment helps to identify problems and may be essential in chronic and disabling diseases such as in Parkinson’s disease (PD). There are numerous reasons for the decrease in QoL suffered by PD patients: restrictions in mobility, falls, emotional disorders, social embar-

assment, isolation, sleep disturbances, dyskinesias, and fluctuations. Many aspects of these disorders go unnoticed in clinical evaluation, and only QoL assessment allows them to be rated. Generic instruments have been used in a few studies measuring QoL in PD patients. Only recently have specific instruments as PDQ-39 and PDQL-37 been designed and validated. QoL assessment in PD is an important and expanding area, with a promising application in clinical trials and pharmacoeconomics.

Key words Quality of life · Parkinson’s disease · Assessment · Questionnaires · Cost-effectiveness

Introduction

The concept of “quality of life” (QoL) is considered complementary to that of “duration (or quantity) of life”. People want to live longer, but they also want to live better. The sense of well-being is the core of the QoL concept, but the latter has many additional components which are quite variable. Some of these are related mainly to individual traits such as health, vocation, and spiritualism; others depend on family or society. The possibility of interactions between all these factors is evident and is therefore recognized in the definition of health as proposed by the World Health Organization: “Health is a state of complete physical, mental and social well-being and not merely the absence of disease or infirmity” [26].

This contribution refers to specifically health-related QoL. Spilker [22] has noted that “While it is widely agreed that health-related quality of life is not the only type of quality of life, there is no agreement on the number or identity of other specific types”.

No definition of QoL has yet gained universal acceptance. From a pragmatic point of view, QoL in clinical medicine refers to the patient’s own perception and self-evaluation regarding the effects of an illness and its consequences on her or his life. QoL assessment is therefore subjective, individual, multidimensional, self-administered, and temporally variable [19]. The most relevant aspects to be considered in QoL are physical status and functional ability, psychological status, social interactions, economic and vocational status, and religious and spiritual status. Some QoL instruments are limited to three or four dimensions since economic and religious sta-

tus may be considered as unrelated to the objectives of the study, but it is nevertheless important to make evaluations as complete as possible. The only exception is the religious aspect, due to its complexity and because it may be too deeply rooted and thus would not be modified either by disease or by therapy. However, even these arguments may be challenged [22].

The interest in QoL assessments stems from the subjective perception of health. Two relevant aspects are linked to this perception: the demand of health care resources and the possibility of disagreement with the clinical evaluation. Both items justify, along with a number of others, the inclusion of QoL measurement in clinical trials. Furthermore, an adequate evaluation of the outcomes of research programmes requires analysis of efficacy, cost effectiveness and net benefit for the patient in order to balance costs and benefits [23]; QoL assessment is therefore of paramount importance to this goal. In addition, QoL assessment provides the basis for comparing various disease effects to identify problems (thus allowing the search and implementation of solutions) and to distinguish between therapies whose differences in efficacy and/or safety are only marginal. Finally, QoL improvement could be the main endpoint in some clinical trials, such as in those carried out in palliative care or incurable diseases [8].

Quality of life measures

Methods for the measurement of QoL are classified as generic or specific depending on the population and health conditions which to be investigated.

Generic instruments include health profiles and utility measures. Health profiles are designed as questionnaires that attempt to assess the most important aspects of QoL. Generic instruments are multidimensional, cover a variety of areas and can be applied to different health states and diseases. These tools are considered more appropriate for studies on health care programmes, surveys of general health status, and comparisons between diseases. Their potential drawbacks are the lack of sensitivity in some areas of interest and their low responsiveness to specific changes [9]. Some of the most frequently used generic health assessment profiles are the Sickness Impact Profile [2], the Nottingham Health Profile [10] and the Short-Form Health Survey (SF-36) [25].

Utility measures are generic instruments characterized by preference measurements and references to full health or death. A numerical score, referred to as “utility” (deriving from cost-utility analysis), is assigned to the QoL status, with a value between 0.0 (death) and 1.0 (full health). The major disadvantages of utility measures are similar to those for health profiles. A typical example of this kind of instrument is the visual analog scale with scores ranking from 0 for the least desirable status (death) to 100 for the most desirable (perfect health).

Specific instruments focus on the aspects considered most relevant to the respective area of interest. These may be specific for a given disease, population, function or condition. Their main advantages derive from the prevalence and responsiveness of areas under assessment and are thus more appropriate than generic instruments for clinical trials. The evaluation of restricted domains and the limitation to specific populations are potential disadvantages of these instruments [9]. Examples of this type of instrument worth mentioning are the Arthritis Impact Measurement Scales [16], the Dyspnea Index [14], and the Epilepsy Surgery Inventory [24].

The following are some of the most relevant steps in assessing QoL (modified from [17]):

- Define the purpose of the assessment
- Describe the population
- Conceptualize QoL outcomes
- Choose QoL measures to be applied
- Carry out the pretest or pilot study
- Collect data
- Analyse outcomes

The development and validation of a QoL measure require methods to check the acceptability, reliability, validity, responsiveness and interpretability of the instrument [12, 13]. A set of criteria have been proposed for development of a QoL questionnaire: it must reflect areas which are important for patients; both physical and emotional health should be measured; summary scores should be amenable to statistical analysis; it should be valid, relatively short, and simple [4]. In addition, the administration method must be defined, both in terms of the person performing the measurement (self-administered, proxy, observer) and means for carrying it out (face-to-face interview, by phone, postal survey). The length of the instrument, the cost of administering it, the methods of scoring and analysis of data, and the presentation and interpretability of results must also be considered and stipulated [17].

Quality of life in Parkinson's disease

What, how and why to measure are key issues in evaluating QoL in Parkinson's disease (PD). Some answers to these questions may emerge from the present review, but prior to this a basic statement must be noted. The clinical evaluation of PD is usually carried out using scales and simple objective tests (timed tests). The information obtained from includes a variety of functional and mental aspects, in addition to data gathered in the patient's examination. However, many factors that are important to the patient may remain unassessed. Emotional and psychosocial consequences of the disease are usually omitted from the physician's evaluation. The frustrations and restrictions caused by the disease are heavy burdens for patients

Table 1 Studies applying QoL measures in Parkinson's disease

	Stebbins et al. [27]	Baron et al. [1]	McRae et al. [15]	Block et al. [3]
Treatment	Implant Adrenal medulla (autograft)	Pallidotomy	Implant Dopamine cells (embryonic)	FIRST study*
Number of patients	7	15	20	618 (randomized, two groups)
Duration	2 years	1 year	2 years	5 years
Instruments	Sickness Impact Profile	Medical Outcomes Study	Battery of scales	Nottingham Health Profile
Main findings	1 year – Improvement – Sleep and rest – Social isolation – Ambulation 2 year – Loss of the improvement	Improvement in: – Physical role – Social functioning – Vitality	6 and 12 months: – More anxiety – Less social support than norm groups 24 months: – Decrease of anxiety	At end of year 5, statistical difference favoring Sinemet® CR in: – Emotional reaction – Social isolation

* First Study: Comparison of immediate-release and controlled release carbidopa/levodopa [3]

and their families. Clinical rating may provide data on severity of the disease or on the degree of disability, but in no way are the medical evaluations equivalent to the QoL status perceived by the patient.

There are many sources for distress and QoL decrease in PD. These include: gait disorder and slowness, freezing, falls, troubles in manual ability for activities such as feeding, dressing and bathing, constipation, drooling, dysphagia, depression, social embarrassment, communication problems, sleep disorders, fatigue, painful spasms, withdrawal, isolation and loneliness, loss of hobbies and leisure activities, driving inability, severe dyskinesias, hallucinations and delirium related to therapy with dopaminergic drugs, and loss of employment in case of active patients [6, 21]. The obvious consequence of this plethora of problems is a decrease in QoL, but the question is whether investigators and clinicians have adequate tools for quantifying all these troubles. A number of studies of PD have been performed in recent years that have applied generic QoL measures (Table 1).

Stebbins et al. [27] studied seven PD patients who underwent implant surgery with adrenal medullary autograft. The Sickness Impact Profile was applied to measure changes in QoL. The authors reported improvement in QoL aspects such as sleep and rest, social isolation and ambulation 1 year after surgery, and stated that some of these changes would have remained unnoticed in the absence of an appropriate QoL evaluation. The improvement in QoL was well correlated with improvement in motor function, as assessed by the Schwab and England Scale [20], and disappeared 2 years after surgery, paralleling the decline in motor function.

Baron et al. [1] reported the outcome of posterior internal pallidotomy in 15 patients with medically intractable PD. A battery of scales and tests for evaluating

patients' physical and mental functioning were applied preoperatively and at various intervals after surgery. In addition, the eight scales of the Medical Outcomes Study were administered to seven nondemented patients preoperatively, early after surgery and 1 year postoperatively. Significant improvements were recorded in tremor, rigidity, akinesia and gait dysfunction along with a reduction in dyskinesias and fluctuations. Physical and social functioning and vitality dimensions of the Medical Outcomes Study showed significant improvement after surgery in spite of the mean total Unified Parkinson's Disease Rating Scale (UPDRS) [5], and the Schwab and England Scale revealed no statistically significant decline.

McRae et al. [15] investigated QoL and psychosocial adjustment in 20 PD patients receiving stereotactic implantation of human embryonic dopamine cells. Measure questionnaires (scales and health indices, such as the Social Provisions Scale and the UCLA Loneliness Scale) were applied to patients and care-givers, as appropriate. It was found that both patients and caregivers experienced "significantly more anxiety and less social support than norm groups at both 6 and 12 months ($p < 0.05$)". A very interesting finding was that the physician's assessment of PD stage at 6 months is predictive of patient reports of depression, anxiety, global health, and life satisfaction 1 year after surgery.

Finally, Block et al. [3] reported on a 5-year, multicentric and comparative study in 618 PD patients who had not previously been exposed to levodopa therapy. The patients were randomized for treatment with carbidopa/levodopa standard (Sinemet 25/100) or with a carbidopa/levodopa sustained-release formulation (Sinemet CR 50/200). The authors applied clinical rating scales and the Nottingham Health Profile. During the study the activities of UPDRS section on activities of daily living consis-

tently favoured the group treated with Sinemet CR. Concerning the Nottingham Health Profile, a statistically significant difference ($p < 0.05$) in the dimensions of emotional reaction and social isolation was recorded in favour of Sinemet CR patients at the end of year 5.

It is now widely accepted that it is more convenient to use specific instruments for specific diseases, especially in clinical trials. The first specific QoL measure for PD was designed by Jenkinson and co-workers in Oxford [11, 18]. Their questionnaire, named PDQ-39, consists of 39 items included in eight dimensions. A second measure was published by de Boer and co-workers in 1996 [4] as PDQL and includes 37 items in four dimensions: parkinsonian symptoms, systemic symptoms, social functioning, and emotional functioning. Studies on the validity of these instruments in countries different to those of origin are needed to check the influence of ethnic, cultural, and linguistic peculiarities.

As in other chronic and disabling diseases, it may also be very interesting to evaluate the QoL of caregivers. Glozman and co-workers [7] presented a report on this aspect of PD in 1997, stressing the relationship between care-givers' QoL, the level of care of the patient, and the degree of disability of the patient.

Conclusions

QoL measurement can be carried out in PD patients, and it is very convenient. Four points may be highlighted to summarize the arguments presented here: (a) QoL assessment contributes to a better understanding of the disease's consequences, and its treatment, on the patient, and thus helps in decision making; (b) QoL evaluations reflect the point of view of patients (who may disagree with the clinical ratings); (c) although usually correlated, QoL may not be equated with disability; (d) many relevant aspects related to the emotional and psychosocial well-being of patients cannot be evaluated appropriately by clinical methods.

Some authors have applied generic QoL measures, such as the Sickness Impact Profile and the Nottingham Health Profile, in clinical trials involving surgical or drug therapy (Table 1). At present a number of valid and specific instruments are available (PDQ-39, PDQL-37) [4, 18] for assessing QoL in PD. QoL measures might become essential in comprehensive evaluations of PD patients as well as in performing adequate cost-benefit analysis in clinical trials and in allocation of resources.

References

- Baron MS, Vitek JL, Bakay RAE, Green J, Kaneoke Y, Hashimoto T, Turner RS, Woodard JL, Cole SA, McDonald WM, DeLong MR (1996) Treatment of advanced Parkinson's disease by posterior Gpi pallidotomy: 1-year results of a pilot study. *Ann Neurol* 40:355-366
- Bergner M, Bobbitt RA, Pollard WE, Martin DP, Gilson BS (1976) The Sickness Impact Profile: validation of a health status measure. *Med Care* 14:57-67
- Block G, Liss C, Reines S, Irr J, Nibbelink D, the CR First Study Group (1997) Comparison of immediate-release and controlled release carbidopa/levodopa in Parkinson's disease. A multicentric 5-year study. *Eur Neurol* 37:23-27
- de Boer AGEM, Wijker W, Speelman JD, de Haes JCJM (1996) Quality of life in patients with Parkinson's disease: development of a questionnaire. *J Neurol Neurosurg Psychiatry* 61:70-74
- Fahn S, Elton RL, Members of the UPDRS Development Committee (1987) Unified Parkinson's Disease Rating Scale. In: Fahn S, Marsden CD, Calne DB, Goldstein M (eds) *Recent developments in Parkinson's disease*, vol 2. Macmillan Health Care Information, Florham Park, NJ, pp 153-164
- Fitzsimmons B, Bunting LK (1993) Parkinson's disease. Quality of life issues. *Nursing Clin N Am* 28:807-818
- Glozman JM, Bitcheva KG, Fedorova NV (1997) Parkinson's disease and quality of life of caregivers. *Mov Disord* 12 [Suppl 1]:48
- Gotay CC, Korn EL, McCabe MS, Moore TD, Cheson BD (1992) Quality-of-life assessment in cancer treatment protocols: research issues in protocol development. *J Natl Cancer Inst* 84:575-579
- Guyatt GH, Jaeschke R, Feeny DH, Patrick DL (1996) Measurements in clinical trials: choosing the right approach. In: Spilker B (ed) *Quality of life and pharmacoeconomics in clinical trials*. Lippincott-Raven, Philadelphia, pp 41-48
- Hunt SM, McEwen J, McKenna SP (1986) *Measuring health status*. Croom Helm, London
- Jenkinson C, Peto V, Fitzpatrick R, Greenhall R, Hyman N (1995) Self-reported functioning and well-being in patients with Parkinson's disease: comparison of the Short-form Health Survey (SF-36) and the Parkinson's Disease Questionnaire (PDQ-39). *Age Ageing* 24:505-509
- Juniper EF, Guyatt GH, Jaeschke R (1996) How to develop and validate a new health-related quality of life instrument. In: Spilker B (ed) *Quality of life and pharmacoeconomics in clinical trials*. Lippincott-Raven, Philadelphia, pp 49-56
- Kirshner B, Guyatt G (1985) A methodological framework for assessing health indices. *J Chron Dis* 38:27-36
- Mahler DA, Weinberg DH, Wells CK, Feinstein AR (1984) Measurement of dyspnea. Contents, interobserver correlates of two new clinical indexes. *Chest* 85:751-758
- McRae C, O'Brien C, Freed C (1996) Quality of life among persons receiving neural implant surgery for Parkinson's disease. *Mov Disord* 11:605-606

-
16. Meenan RF, Gertman PM, Mason JH (1980) Measuring health status in arthritis: the Arthritis Impact Measurement Scales. *Arthritis Rheum* 23: 146–152
 17. Patrick DL, Erickson P (1993) Assessing health-related quality of life for clinical decision-making. In: Walker SR, Rosser RM (eds) *Quality of life assessment: key issues in the 1990s*. Kluwer, Dordrecht, pp 10–63
 18. Peto V, Jenkinson C, Fitzpatrick R, Greenhall R (1995) The development and validation of a short measure of functioning and well being for individuals with Parkinson's disease. *Qual Life Res* 4: 241–248
 19. Schipper H, Clinch JJ, Olweny CLM (1996) Quality of life studies: definitions and conceptual issues. In: Spilker B (ed) *Quality of life and pharmacoeconomics in clinical trials*. Lippincott-Raven, Philadelphia, pp 11–23
 20. Schwab RS, England AC (1969) Projection technique for evaluating surgery in Parkinson's disease. In: Gillingham FJ, Donaldson IML (eds) *Third symposium on Parkinson's disease*. Livingstone, Edinburgh, pp 152–157
 21. Shindler JS, Brown R, Welburn P, Parkes JD (1993) Measuring the quality of life of patients with Parkinson's disease. In: Walker SR, Rosser RM (eds) *Quality of life assessment: key issues in the 1990s*. Kluwer, Dordrecht, pp 289–300
 22. Spilker B (1996) Introduction. In: Spilker B (ed) *Quality of life and pharmacoeconomics in clinical trials*. Lippincott-Raven, Philadelphia, pp 1–10
 23. Testa MA, Simonson DC (1996) Assessment of quality-of-life-outcomes. *New Engl J Med* 334: 835–840
 24. Vickrey BG, Hays RD, Graber J, Rausch R, Engel J, Brook RH (1992) A health-related quality of life instrument for patients evaluated for epilepsy surgery. *Med Care* 30: 299–319
 25. Ware JE, Sherbourne CD (1992) The MOS 36-item Short-Form Health Survey (SF-36). I. Conceptual framework and item selection. *Med Care* 30: 473–483
 26. World Health Organization (1952) *Handbook of basic documents*, 5th edn. Palais des Nations, Geneva, pp 3–20
 27. Wilson RS, Goetz CG, Stebbins GT (1996) Neurologic illness. In: Spilker B (ed) *Quality of life and pharmacoeconomics in clinical trials*. Lippincott-Raven, Philadelphia, pp 903–908